Relations and functions - a quick review

Relations

A relation $R(a, b, \ldots, z)$, where $a \in A,\ b \in B,\ \ldots, z \in Z$, is a subset of the cartesian product $A \times B \times \ldots \times Z$. The product $A \times B \times \ldots \times Z$ is the domain of R.

If $\langle a, b, \ldots, z \rangle \in R$, we say that R holds on $\langle a, b, \ldots, z \rangle$, that $\langle a, b, \ldots, z \rangle$ satisfies R, or that $R(a, b, \ldots, z)$ is true.

Functions

A function $f(x) = y$, where $x \in X$ and $y \in Y$ maps elements of the domain X to elements of the range or codomain Y.

A function f is injective or one-to-one if whenever $a \neq b$, then $f(a) \neq f(b)$.

A function f is surjective or onto if for every element y of the range Y, there exists some x such that $f(x) = y$.

A function f is bijective if it is both injective and surjective.

Relations and functions

A function $f(x)$ is equivalent to a relation R that holds only for the pairs $\langle x, f(x) \rangle$.

A relation $R(a, b, \ldots, z)$ can be viewed as a function $f(a, b, \ldots, z)$ that results in the value True when $R(a, b, \ldots, z)$ holds, and False otherwise.

If we have a tuple $t = \langle t_0, \ldots, t_{n-1} \rangle$, we can talk about both relations and functions in unary or n-ary forms: $R(t)$ vs. $R(t_0, \ldots, t_{n-1})$. For relations, the n-ary form is more common. For functions, the unary form is more common in mathematics, while the n-ary form is more common in programming.